

CPU Architecture & Register Organization

Q. Explain the internal organization of a CPU. Describe the basic components including Registers, ALU, and the Control Unit.

[DEF] Definition to Remember

The CPU (Central Processing Unit) is the brain of the computer. Its internal architecture defines how it fetches, decodes, and executes instructions. It consists of three primary components interconnected by an internal bus: Registers (storage), ALU (processing), and the Control Unit (management).

1. Registers

High-speed, temporary storage locations situated directly on the CPU chip.

* Ge

2. Arithmetic and Logic Unit (ALU)

The execution core where actual data processing happens.

* Ari

3. Control Unit (CU)

The manager of the CPU. It does not process data itself.

* Fu

4. Internal CPU Organization

The Registers, ALU, and CU are connected via an Internal CPU Bus. This localized data pathway ensures high-speed data transfer between CPU components without having to use the slower external system bus.

[MUST-WRITE] Must-Write Points (for 10 marks)

1. CPU has 3 main components: Registers, ALU, and Control Unit (CU).
2. Registers: Extremely fast, on-chip storage for temporary data and addresses.
3. PC (Program Counter) stores next instruction address; IR holds current instruction.
4. MAR/MBR handle memory addresses/data; Accumulator stores math results.
5. ALU: Performs all Arithmetic (add, sub) and Logical (AND, OR) operations.
6. Control Unit (CU): Decodes instructions and issues control signals to direct hardware.
7. Components communicate over high-speed Internal CPU Buses.

[RECALL] Quick Recall

CPU Org -> Registers (PC/IR/MAR/MBR/ACC) -> ALU (Math + Logic) -> CU (Manager/Signals) -> Internal Bus connects them